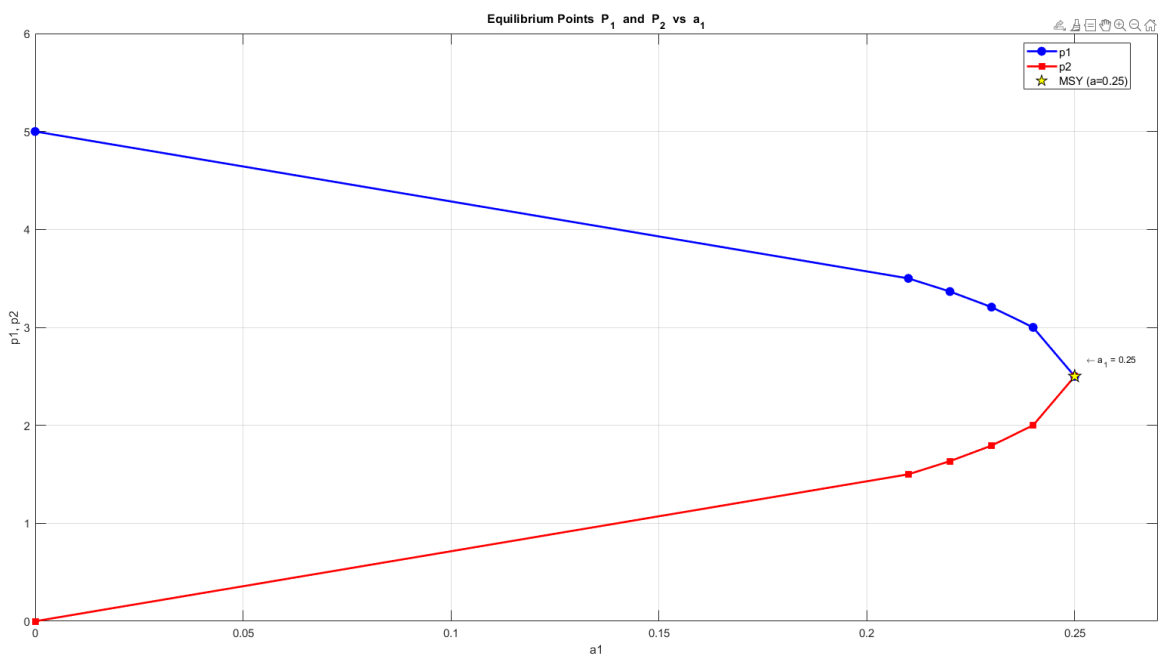
```
k  = 0.20;
N  = 5;
a1 = 0.21;
a2 = 0.25;   % MSY = k*N/4 = 0.25
b  = 1;

%% ==========================================================
%  FIGURE 1 - Equilibrium points P1 & P2 vs a1
a_vals  = [0,    0.21,         0.22,         0.23,         0.24, 0.25];
P1_vals = [5,    3.5,          3.366025404,   3.207106781,   3,    2.5];
P2_vals = [0,    1.5,          1.633974596,   1.792893219,   2,    2.5];

figure(1);
set(gcf, 'Name', 'Figure 1 - Equilibrium Points vs a1', ...
        'NumberTitle', 'off', 'Color', 'w', 'Position', [80 500 560 400]);

plot(a_vals, P1_vals, 'b-o', 'LineWidth', 2, 'MarkerSize', 7, ...
    'MarkerFaceColor', 'b', 'DisplayName', 'p1');
hold on;
plot(a_vals, P2_vals, 'r-s', 'LineWidth', 2, 'MarkerSize', 7, ...
    'MarkerFaceColor', 'r', 'DisplayName', 'p2');

% Highlight MSY meeting point
plot(0.25, 2.5, 'kp', 'MarkerSize', 13, 'MarkerFaceColor', 'y', ...
    'DisplayName', 'MSY (a=0.25)');
```

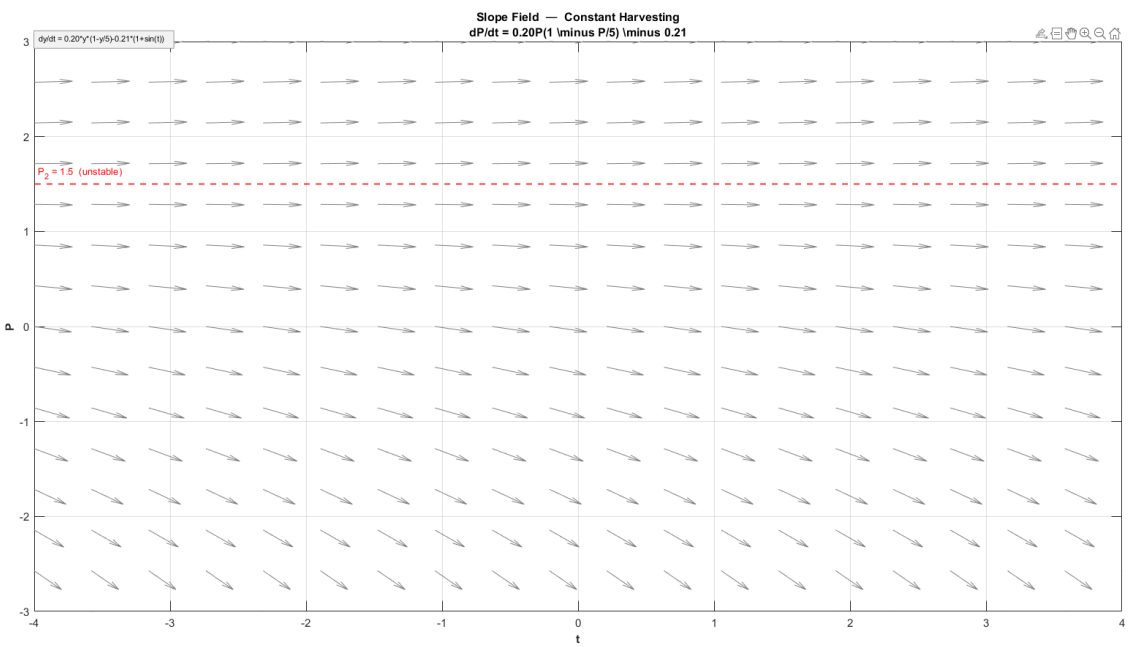
```
xlabel('a1', 'FontSize', 13);

ylabel('p1, p2', 'FontSize', 13);

title('Equilibrium Points  P_1  and  P_2  vs  a_1', 'FontSize', 13, 'FontWeight', 'bold');

legend('p1', 'p2', 'MSY (a=0.25)', 'Location', 'best', 'FontSize', 11);

xlim([0 0.27]);

ylim([0 6]);

grid on;

set(gca, 'FontSize', 11, 'Box', 'on');

text(0.252, 2.65, ' \leftarrow a_1 = 0.25', 'FontSize', 9);
```




```
%  Equilibrium values for a = a1 = 0.21

disc1 = N^2 - 4*(a1*N/k);

P1_eq = (N + sqrt(disc1)) / 2;   % = 3.5

P2_eq = (N - sqrt(disc1)) / 2;   % = 1.5
```

```
%% ==========================================================
%  FIGURE 2 - Slope field (a = a1 = 0.21)
figure(2);
set(gcf, 'Name', 'Figure 2 - Slope Field Constant Harvest a=0.21', ...
        'NumberTitle', 'off', 'Color', 'w', 'Position', [80 50 640 480]);

% Build slope field grid  (same window as paper: t in [-4,4], P in [-3,3])
[T_sf, P_sf] = meshgrid(linspace(-4, 4, 20), linspace(-3, 3, 15));
dP_sf = k .* P_sf .* (1 - P_sf./N) -  a1 .* (1 + sin(b .* T_p));
dT_sf = ones(size(dP_sf));
mag   = sqrt(dT_sf.^2 + dP_sf.^2);  mag(mag==0) = 1;

quiver(T_sf, P_sf, dT_sf./mag, dP_sf./mag, 0.5, ...
      'Color', [0.5 0.5 0.5], 'LineWidth', 0.7);
hold on;

% Equilibrium horizontal lines
yline(P1_eq, 'b--', sprintf('P_1 = %.1f  (stable)',   P1_eq), ...
     'LineWidth', 1.6, 'LabelHorizontalAlignment', 'left', 'FontSize', 10);
yline(P2_eq, 'r--', sprintf('P_2 = %.1f  (unstable)', P2_eq), ...
     'LineWidth', 1.6, 'LabelHorizontalAlignment', 'left', 'FontSize', 10);

xlabel('t', 'FontSize', 13, 'FontWeight', 'bold');
ylabel('P', 'FontSize', 13, 'FontWeight', 'bold');
title({'Slope Field  —  Constant Harvesting', ...
```

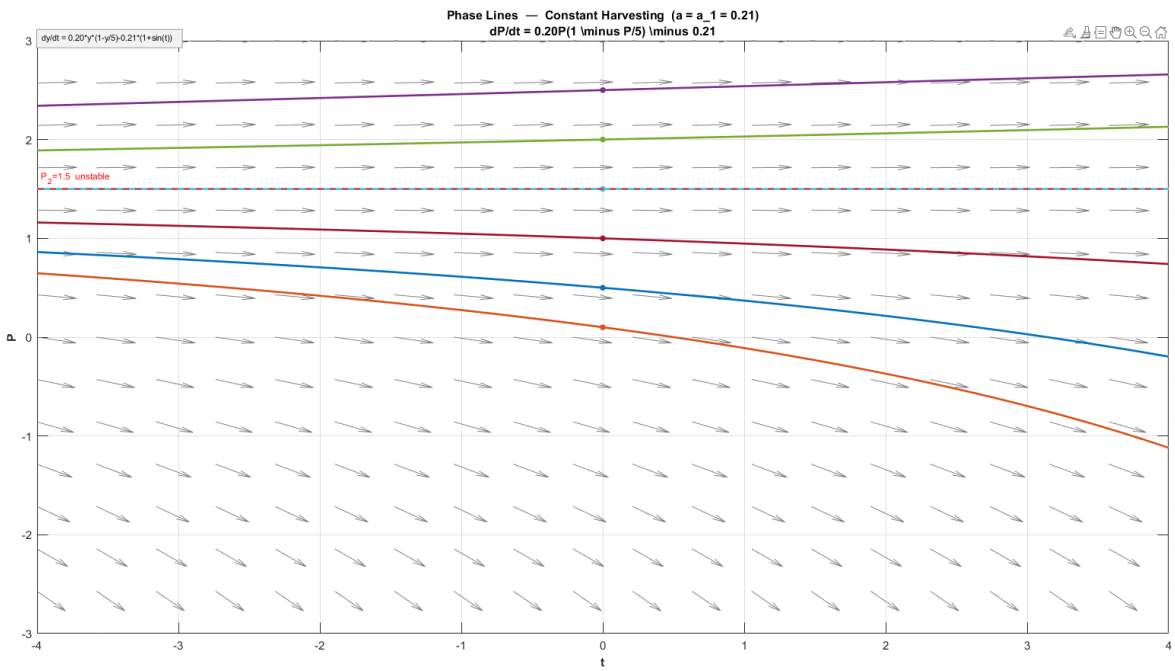
```
     'dP/dt = 0.20P(1 \minus P/5) \minus 0.21'}, ...
    'FontSize', 12, 'FontWeight', 'bold');
xlim([-4 4]);  ylim([-3 3]);
grid on;  set(gca, 'FontSize', 11, 'Box', 'on');

% Header annotation matching DE Tools display in the paper
annotation('textbox', [0.13 0.88 0.76 0.06], ...
   'String', 'dy/dt = 0.20*y*(1-y/5)-0.21*(1+sin(t))', ...
   'FontSize', 8, 'BackgroundColor', [0.95 0.95 0.95], ...
   'EdgeColor', [0.6 0.6 0.6], 'FitBoxToText', 'on');
```




```
%  FIGURE 2.1 - Phase lines (a = a1 = 0.21, multiple ICs)
figure(3);
set(gcf, 'Name', 'Figure 2.1 - Phase Lines Constant Harvest a=0.21', ...
        'NumberTitle', 'off', 'Color', 'w', 'Position', [740 50 640 480]);
```

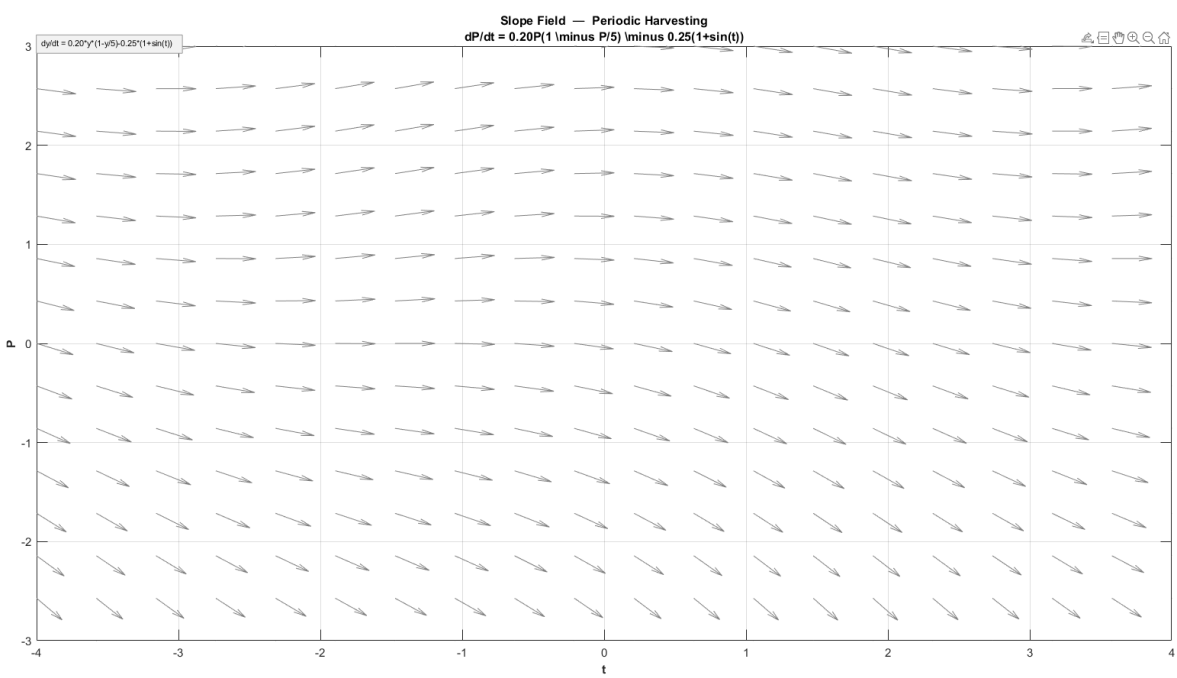
```
% Slope field background
quiver(T_sf, P_sf, dT_sf./mag, dP_sf./mag, 0.5, ...
      'Color', [0.5 0.5 0.5], 'LineWidth', 0.7);
hold on;

% Initial conditions (matching DE Tools IC = 5 as in paper)
ICs   = [5.0, 4.0, 3.5, 2.5, 2.0, 1.5, 1.0, 0.5, 0.1];
clrs  = lines(length(ICs));
opts  = odeset('RelTol', 1e-7, 'AbsTol', 1e-9);

for i = 1:length(ICs)
   P0 = ICs(i);
   c  = clrs(i, :);

   % Integrate forward t: 0 -> 4
   [tf, Pf] = ode45(@(t,P) k*P*(1-P/N) - a1*(1+sin(b*t)), [0  4], P0, opts);
   % Integrate backward t: 0 -> -4
   [tb, Pb] = ode45(@(t,P) k*P*(1-P/N) -  a1*(1+sin(b*t)), [0 -4], P0, opts);

   % Combine & clip
   t_full = [flipud(tb); tf];
   P_full = [flipud(Pb); Pf];
   P_full(P_full < -3.1) = NaN;

   plot(t_full, P_full, '-', 'Color', c, 'LineWidth', 2);
```

```
    plot(0, P0, 'o', 'Color', c, 'MarkerFaceColor', c, 'MarkerSize', 5);
end

% Equilibrium lines
yline(P1_eq, 'b--', sprintf('P_1=%.1f  stable',   P1_eq), ...
      'LineWidth', 1.6, 'LabelHorizontalAlignment', 'left', 'FontSize', 9);
yline(P2_eq, 'r--', sprintf('P_2=%.1f  unstable', P2_eq), ...
      'LineWidth', 1.6, 'LabelHorizontalAlignment', 'left', 'FontSize', 9);

xlabel('t', 'FontSize', 13, 'FontWeight', 'bold');
ylabel('P', 'FontSize', 13, 'FontWeight', 'bold');
title({'Phase Lines  —  Constant Harvesting  (a = a_1 = 0.21)', ...
       'dP/dt = 0.20P(1 \minus P/5) \minus 0.21'}, ...
      'FontSize', 12, 'FontWeight', 'bold');
xlim([-4 4]);  ylim([-3 3]);
grid on;  set(gca, 'FontSize', 11, 'Box', 'on');

annotation('textbox', [0.13 0.88 0.76 0.06], ...
    'String', 'dy/dt = 0.20*y*(1-y/5)-0.21*(1+sin(t))', ...
    'FontSize', 8, 'BackgroundColor', [0.95 0.95 0.95], ...
    'EdgeColor', [0.6 0.6 0.6], 'FitBoxToText', 'on');
```




```
%  FIGURE 3 - Slope field (a = a2 = 0.25)
figure(4);
set(gcf, 'Name', 'Figure 3 - Slope Field Periodic Harvest a=0.25', ...
        'NumberTitle', 'off', 'Color', 'w', 'Position', [80 50 640 480]);

% Slope field — evaluated at t values directly (non-autonomous)
[T_p, P_p] = meshgrid(linspace(-4, 4, 20), linspace(-3, 3, 15));
dP_p = k .* P_p .* (1 - P_p./N) - a2 .* (1 + sin(b .* T_p));
dT_p = ones(size(dP_p));
mag_p = sqrt(dT_p.^2 + dP_p.^2);  mag_p(mag_p==0) = 1;

quiver(T_p, P_p, dT_p./mag_p, dP_p./mag_p, 0.5, ...
      'Color', [0.5 0.5 0.5], 'LineWidth', 0.7);
hold on;
```

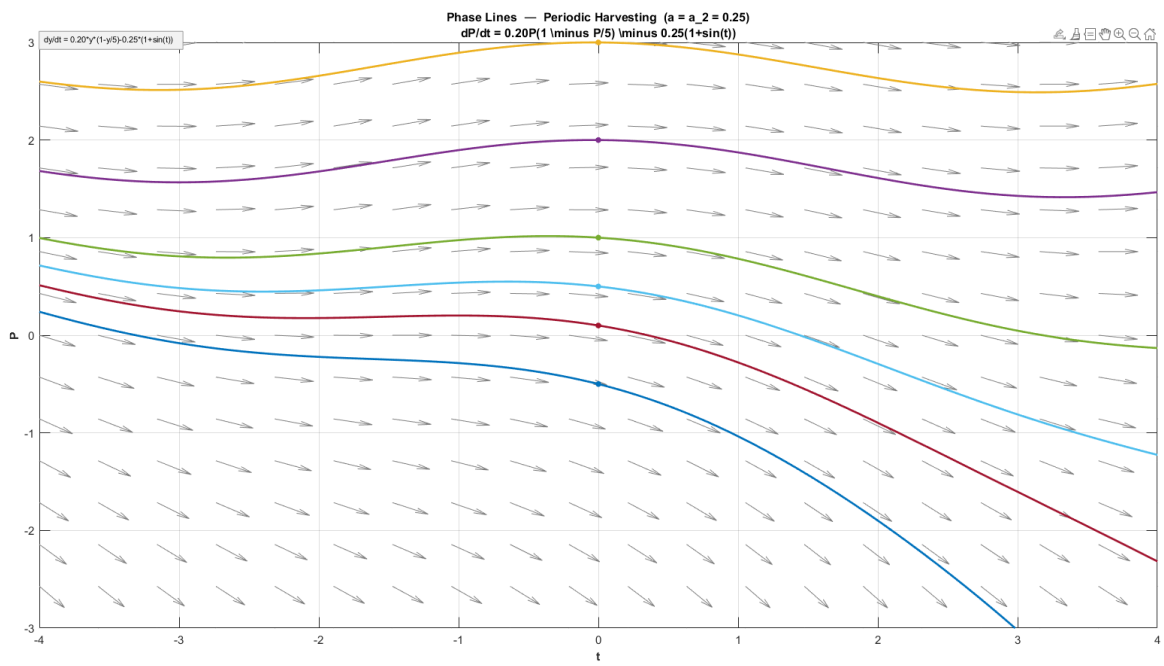
```
xlabel('t', 'FontSize', 13, 'FontWeight', 'bold');
ylabel('P', 'FontSize', 13, 'FontWeight', 'bold');
title({'Slope Field  —  Periodic Harvesting', ...
      'dP/dt = 0.20P(1 \minus P/5) \minus 0.25(1+sin(t))'}, ...
     'FontSize', 12, 'FontWeight', 'bold');
xlim([-4 4]);  ylim([-3 3]);
grid on;  set(gca, 'FontSize', 11, 'Box', 'on');

annotation('textbox', [0.13 0.88 0.76 0.06], ...
   'String', 'dy/dt = 0.20*y*(1-y/5)-0.25*(1+sin(t))', ...
   'FontSize', 8, 'BackgroundColor', [0.95 0.95 0.95], ...
   'EdgeColor', [0.6 0.6 0.6], 'FitBoxToText', 'on');
```




```
%  FIGURE 3.1 - Phase lines (a = a2 = 0.25, multiple ICs)
figure(5);
```

```
set(gcf, 'Name', 'Figure 3.1 - Phase Lines Periodic Harvest a=0.25', ...
        'NumberTitle', 'off', 'Color', 'w', 'Position', [740 50 640 480]);

quiver(T_p, P_p, dT_p./mag_p, dP_p./mag_p, 0.5, ...
      'Color', [0.5 0.5 0.5], 'LineWidth', 0.7);
hold on;

ICs2  = [5.0, 4.0, 3.0, 2.0, 1.0, 0.5, 0.1, -0.5];
clrs2 = lines(length(ICs2));

for i = 1:length(ICs2)
   P0 = ICs2(i);
   c  = clrs2(i, :);

   [tf2, Pf2] = ode45(@(t,P) k*P*(1-P/N) - a2*(1+sin(b*t)), [0  4], P0, opts);
   [tb2, Pb2] = ode45(@(t,P) k*P*(1-P/N) - a2*(1+sin(b*t)), [0 -4], P0, opts);

   t_full2 = [flipud(tb2); tf2];
   P_full2 = [flipud(Pb2); Pf2];
   P_full2(P_full2 < -3.1) = NaN;   % clip below window

   plot(t_full2, P_full2, '-', 'Color', c, 'LineWidth', 2);
   plot(0, P0, 'o', 'Color', c, 'MarkerFaceColor', c, 'MarkerSize', 5);
end

xlabel('t', 'FontSize', 13, 'FontWeight', 'bold');
```

```
ylabel('P', 'FontSize', 13, 'FontWeight', 'bold');
title({'Phase Lines  —  Periodic Harvesting  (a = a_2 = 0.25)', ...
      'dP/dt = 0.20P(1 \minus P/5) \minus 0.25(1+sin(t))'}, ...
     'FontSize', 12, 'FontWeight', 'bold');
xlim([-4 4]);  ylim([-3 3]);
grid on;  set(gca, 'FontSize', 11, 'Box', 'on');


annotation('textbox', [0.13 0.88 0.76 0.06], ...
   'String', 'dy/dt = 0.20*y*(1-y/5)-0.25*(1+sin(t))', ...
   'FontSize', 8, 'BackgroundColor', [0.95 0.95 0.95], ...
   'EdgeColor', [0.6 0.6 0.6], 'FitBoxToText', 'on');
```




```
%   print Table 1 values for verification
fprintf('\n========== TABLE 1  (Choice 3: k=%.2f, N=%d) ==========\n', k, N);
```

```
fprintf('%-6s  %-18s  %-18s\n', 'a1', 'P1', 'P2');
fprintf('%s\n', repmat('-',1,46));
a_tbl = [0, 0.21, 0.22, 0.23, 0.24, 0.25];
for i = 1:length(a_tbl)
   ai = a_tbl(i);
   d  = N^2 - 4*(ai*N/k);
   if d >= 0
      fprintf('%-6.2f  %-18.9f  %-18.9f\n', ai, ...
            (N+sqrt(d))/2, (N-sqrt(d))/2);
   else
      fprintf('%-6.2f  %-18s  %-18s\n', ai, 'undefined', 'undefined');
   end
end
fprintf('%s\n', repmat('=',1,46));
```

```
=========== TABLE 1  (Choice 3: k=0.20, N=5) ===========
a1     P1                P2
----------------------------------------------
0.00   5.000000000       0.000000000
0.21   3.500000000       1.500000000
0.22   3.366025404       1.633974596
0.23   3.207106781       1.792893219
0.24   3.000000000       2.000000000
0.25   2.500000000       2.500000000
==============================================
```